

Maria Camarca

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Education

- 2019 – 2025 **Ph.D. Planetary Science**
California Institute of Technology, Pasadena CA
- 2019 – 2021 **M.S. Planetary Science**
California Institute of Technology, Pasadena CA
- 2017 – 2019 **Part-time coursework in math & physics**
American University, Washington, D.C.
- 2013 – 2017 **B.S. Biology & minor in Psychology**
Marymount University, Arlington VA

Research Positions

- Fall 2026* **NSF Astronomy & Astrophysics Postdoctoral Fellow**
National Radio Astronomy Observatory, Charlottesville VA
- 2025 – 2026 **Postdoctoral Scholar Research Associate**
California Institute of Technology, Pasadena CA
- 2021 – 2025 **NSF Graduate Research Fellow**
- 2019 – 2021 **Graduate Student**
California Institute of Technology, Pasadena CA
- 2017 – 2019 **Postbaccalaureate Research Assistant**
NASA Goddard Space Flight Center | American University
- 2017 **Undergraduate Research Associate in Astrobiology**
NASA Goddard Space Flight Center | American University

Select Honors

- 2025 Caltech Graduate Studies Office Excellence in Leadership and Service Award
- 2022 – 2026 Keck Institute for Space Studies Affiliate
- 2013 – 2017 Marymount University Presidential Scholarship
- 2016 Virginia Foundation for Independent Colleges Fellowship

Publications

- First-author 5. **Camarca, M.**, et al. 2026. “Callisto from JWST: CO₂-rich terrain on the leading hemisphere and global patterns of H₂O ice”. *The Planetary Science Journal*, in review.
4. **Camarca, M.**, et al. 2025. “[A Multifrequency Global View of Callisto’s Thermal Properties from ALMA](#)”. *The Planetary Science Journal*, Vol. 6, No. 8.
3. **Camarca, M.**, et al. 2023. “[Thermal Properties of the Leading Hemisphere of Callisto Inferred from ALMA Observations](#)”. *The Planetary Science Journal*, Vol. 4, No. 8.
2. **Camarca, M.**, Murphy, B., and Jaber, D. 2022. “[The Evolution of Games and Case Studies in Organic Chemistry](#)”. *The Chemical Educator*, Vol. 27, No. 1.
1. **Camarca, M.**, Heuett, W., and Jaber, D. 2019. “[CHEMCompete-II: An Organic Chemistry Card Game To Differentiate between Substitution and Elimination Reactions of Alcohols](#)”. *Journal of Chemical Education*, Vol. 96, No. 11.

- Co-author
11. Bézard, B., [...], **Camarca, M.**, et al. 2026. “An unidentified absorption feature at 5.11 μm on the surface of Titan and Pluto from JWST spectroscopy”. *Astronomy and Astrophysics*, submitted.
 10. Meyer, C., **Camarca, M.**, et al. 2026. “[Revealing Callisto’s Near-subsurface Thermophysical Properties with ALMA Calibration Data](#)”. *The Planetary Science Journal*, Vol. 7, No. 1.
 9. Milby, Z., de Kleer, K., Schmidt, C., and **Camarca, M.** 2026. “[Variability of Europa’s Optical Aurora on Orbital and Multiyear Timescales](#)”. *The Planetary Science Journal*, Vol. 7, No. 3.
 8. Cartwright, R. J., [...], **Camarca, M.**, et al. 2024. “[Revealing Callisto’s Carbon-rich Surface and CO₂ Atmosphere with JWST](#)”. *The Planetary Science Journal*, Vol. 5, No. 3.
 7. Thelen, A. E., [...], **Camarca, M.**, et al. 2024. “[Subsurface Thermophysical Properties of Europa’s Leading and Trailing Hemispheres as Revealed by ALMA](#)”. *The Planetary Science Journal*, Vol. 5, No. 2.
 6. de Kleer, K., [...], **Camarca, M.**, et al. 2023. “[The Optical Aurorae of Europa, Ganymede, and Callisto](#)”. *The Planetary Science Journal*, Vol. 4, No. 2.
 5. Pereira, C. L., [...], **Camarca, M.**, et al. 2023. “The two rings of (50000) Quaoar”. *Astronomy and Astrophysics*, Vol. 673.
 4. Faggi, S., [...], **Camarca, M.**, et al. 2021. “[The Extraordinary Passage of Comet C/2020 F3 NEOWISE: Evidence for Heterogeneous Chemical Inventory in Its Nucleus](#)”. *The Astronomical Journal*, Vol. 162, No. 5.
 3. Shepard, M. K., [...], and **Camarca, M.** 2021. “[Asteroid 16 Psyche: Shape, Features, and Global Map](#)”. *The Planetary Science Journal*, Vol. 2, No. 4.
 2. Lippi, M., [...], **Camarca, M. N.**, et al. 2020. “[New Insights into the Chemical Composition of Five Oort Cloud Comets after Re-analysis of Their Infrared Spectra](#)”. *The Astronomical Journal*, Vol. 159, No. 4.
 1. Paganini, L., **Camarca, M. N.**, et al. 2019. “[Observations of Jupiter Family Comet 252P/LINEAR During a Close Approach to Earth Reveal Large Abundances of Methanol and Ethane](#)”. *The Astronomical Journal*, Vol. 158, No. 3.

Recent Presentations

Feb 2026	Pomona College Physics Colloquium Claremont CA
Sep 2025	DPS-EPSC Joint Meeting (talk) Helsinki, Finland
Aug 2025	JHU-APL Brown Bag Seminar Seminar (invited) Laurel MD
Oct 2024	56th AAS Division of Planetary Science meeting (talk) Boise ID
Oct 2023	55th AAS Division of Planetary Science meeting (talk) San Antonio TX
Jan 2023	NRAO/UVA Tuna Talk (invited) Charlottesville, VA
Oct 2022	54th AAS Division of Planetary Science meeting (talk) London, Ontario
Oct 2021	53rd AAS Division of Planetary Science meeting (virtual poster)

Telescope Programs

ALMA	2 co-I programs: Cycle 11 PI: A. Akins, Cycle 11 PI: A. Thelen
JWST	1 co-I program: Cycle 3 (MIRI) PI: R. Davis
Palomar	1 co-I program: 2022A (CHIMERA) PI: K. de Kleer
Keck	1 Science-PI program: 2020B (NIRSPEC) PI: G. Blake 2 co-I programs: 2021B, 2022A (HIRES) PI: K. de Kleer

Teaching Experience

2021 – 2024	Teaching Assistant Caltech Planetary Science Department Ge 103/11c: Introduction to the Solar System
2021 – 2022	RISE Program Tutor Caltech Y Elementary and high school math and biology
2021 – 2022	Supplemental Instructor/General Tutor Marymount University General and Organic chemistry

Mentoring Experience

2024, 2023	Cole Meyer (Princeton Undergraduate) Caltech SURF Program
2021	Lily Coffin (Caltech Undergraduate) Caltech SURF Program
2021	Tyler Mapes (Caltech Undergraduate) Caltech SURF Program

Select Service and Outreach

Mar 2026	Caltech Astronomy Stargazing Lecture <i>panelist</i>
Apr 2026	Mary W. Jackson Elementary STEAM Night, Altadena CA <i>science demo</i>
2021-2026	Union Station Homeless Services, Pasadena, CA <i>10× dinner lead</i>
Aug 2025	Old Town Monrovia Sidewalk Astronomy <i>telescope guide</i>
Sep 2024	Caltech Keck Remote Observing Room <i>tour for physics/astronomy teachers</i>
Apr 2024	Caltech Astronomy Stargazing Lecture <i>panelist</i>
Feb 2024	Astronomy on Tap, Pasadena CA <i>pub outreach talk</i>
Feb 2024	STEAM Night Highland Oaks Elementary, Arcadia CA <i>outreach talk</i>
Jul 2023	Flintridge Center: Youth of Promise program, Pasadena CA <i>outreach talk</i>
Jul 2023	Family Promise Homeless Ministry, St. Philip, Pasadena CA <i>overnight host</i>
2021-2022	Caltech Dix Planetary Science Seminar <i>co-organizer</i>